

Relational Schema

Note: All foreign keys' update rules are ON UPDATE NO ACTION.

-- PEOPLE (hierarchy H1: ACCOUNTS -> CUSTOMERS, STAFF)

ACCOUNTS (account_id PK, email UQ, username UQ, password_hash, created_at)

CUSTOMERS (customer_id PK FK → ACCOUNTS.account_id [ON DELETE CASCADE], first_name, last_name, phone, is_active, created_at, updated_at)

STAFF (account_id PK FK → ACCOUNTS.account_id [ON DELETE CASCADE], employee_no UQ, department, hire_date, is_active)

ADDRESSES (address_id PK, account_id FK → CUSTOMERS.customer_id [ON DELETE CASCADE], line1, line2, city, region, postal_code, country_code, is_shipping, is_billing, is_default, created_at, UQ (address_id, account_id))

-- CATALOG

CATEGORIES (category_id PK, name UQ, parent_category_id FK → CATEGORIES.category_id [ON DELETE SET NULL], description)

PRODUCTS (product_id PK, category_id FK → CATEGORIES.category_id [ON DELETE SET NULL], sku UQ, name, description, unit_price, currency, stock_quantity, reorder_level, is_active, created_at, updated_at)

-- VENDORS (hierarchy H3: PAYEE -> CARRIERS, SUPPLIERS)

PAYEE (payee_id PK, payee_type, name UQ, created_at)

CARRIERS (carrier_id PK FK → PAYEE.payee_id [ON DELETE CASCADE], tracking_url_template)

SUPPLIERS (supplier_id PK FK → PAYEE.payee_id [ON DELETE CASCADE], contact_email)

-- SALES (partitioned monthly by order_date)

ORDERS (order_id PK, order_date PK, customer_id FK → CUSTOMERS.customer_id [ON DELETE RESTRICT], (shipping_address_id, customer_id) FK → ADDRESSES (address_id,

account_id) [ON DELETE RESTRICT], (billing_address_id, customer_id) FK → ADDRESSES (address_id, account_id) [ON DELETE RESTRICT], carrier_id FK → CARRIERS.carrier_id [ON DELETE SET NULL], status, currency, tracking_number, subtotal, shipping_cost, tax_amount, total_amount [derived], updated_at)

ORDER_ITEMS (order_item_id PK, order_date PK, (order_id, order_date) FK → ORDERS (order_id, order_date) [ON DELETE CASCADE], product_id FK → PRODUCTS.product_id [ON DELETE RESTRICT], quantity, unit_price, line_total [derived], UQ (order_id, order_date, product_id))

-- PAYMENTS AND REFUNDS (hierarchy H2: PAYMENTS -> CARD, PAYPAL, GIFT_CARD)

PAYMENTS (payment_id PK, order_date PK, (order_id, order_date) FK → ORDERS (order_id, order_date) [ON DELETE RESTRICT], amount, currency, payment_method, processed_at, status, transaction_ref)

CARD_PAYMENTS (payment_id PK, order_date PK, (payment_id, order_date) FK → PAYMENTS (payment_id, order_date) [ON DELETE CASCADE], last4, brand, auth_code)

PAYPAL_PAYMENTS (payment_id PK, order_date PK, (payment_id, order_date) FK → PAYMENTS (payment_id, order_date) [ON DELETE CASCADE], payer_id, payer_email)

GIFT_CARD_PAYMENTS (payment_id PK, order_date PK, (payment_id, order_date) FK → PAYMENTS (payment_id, order_date) [ON DELETE CASCADE], gift_card_id)

REFUNDS (refund_id PK, (payment_id, payment_order_date) FK → PAYMENTS (payment_id, order_date) [ON DELETE RESTRICT], approved_by FK → STAFF.account_id [ON DELETE RESTRICT], amount, currency, reason, status, requested_at, resolved_at)

REFUND_ITEMS (refund_id PK FK → REFUNDS.refund_id [ON DELETE CASCADE], order_item_id PK, order_date PK, (order_item_id, order_date) FK → ORDER_ITEMS (order_item_id, order_date) [ON DELETE RESTRICT], quantity, amount)

-- FINANCE

ACCOUNTS_RECEIVABLE (ar_id PK, (order_id, order_date) FK → ORDERS (order_id, order_date) [ON DELETE RESTRICT], amount_due, currency, due_date, status, created_at)

ACCOUNTS_PAYABLE (ap_id PK, payee_id FK → PAYEE.payee_id [ON DELETE RESTRICT], amount_due, currency, due_date, status)

-- CHATBOT KNOWLEDGE BASE

POLICY_DOCUMENTS (document_id PK, title, version, last_updated, category, file_path, is_active)

POLICY_DOCUMENT_CHUNKS (chunk_id PK, document_id FK → POLICY_DOCUMENTS.document_id [ON DELETE CASCADE], chunk_index, chunk_text, embedding, UQ (document_id, chunk_index))